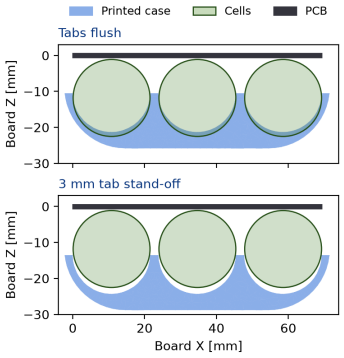
A flush case intersects all three cells before tightening



Flush tabs

Each cell overlaps the printed case by 1.18 mm at its crown. The local nylon-screw solve reached 2.87 of 3.00 mm drawdown. All four tabs crossed the modeled board plane, leaving the physical seating load unresolved.

3 mm stand-off

The unstrained STEP clears the cells and all existing mechanical hardware. Rigid sideways travel of 0.15–0.33 mm closes the first cell gap.

The hole pattern leaves little room for an M4 screw

	PCB	Case
Bore diameter [mm]	4.200	5.000–5.090
Hole pitch X [mm]	62.865	62.855
Hole pitch Y [mm]	114.427	113.241

Each end of the centred pattern is about 0.593 mm off-axis. The smaller bore pair allows 0.600 mm with a nominal M4 shank.

Hardware under study

Four M4 case screws in aluminium or nylon. The nylon reference is McMaster-Carr 93640A145.

The dense PLA case adds about 74 g to a 252 g bare assembly. Print orientation and infill affect its installed stiffness and mass.

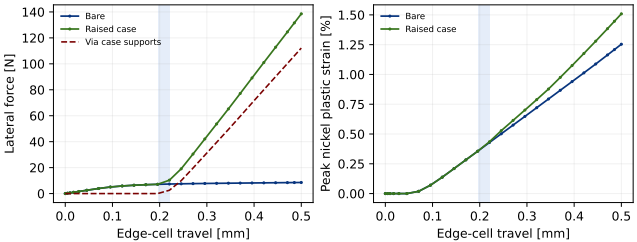
The bare-board preload repeats the October baseline

Bare-board quantity	Value	Loading
First prestressed natural frequency	485.964 Hz	Six bolts locked
Maximum PCB directional displacement	0.052674 mm RMS	Base Z
Maximum PCB equivalent stress	90.325 MPa RMS	Base X

The fresh six-bolt solve held 749.997–749.999 N at each locked fastener and reproduced the October peak static displacement of 0.04746 mm. The fresh native X vibration run also reproduced the October PCB stress of 90.325 MPa RMS.

The fresh bare edge-cell fixture took 8.54 N to reach 0.50 mm sideways travel. The raised case with nylon screws is the first cased comparison.

The raised nylon-screw case restrains the edge cell



The blue band brackets first contact at 0.196–0.221 mm of travel.

At the same 8.54 N pull

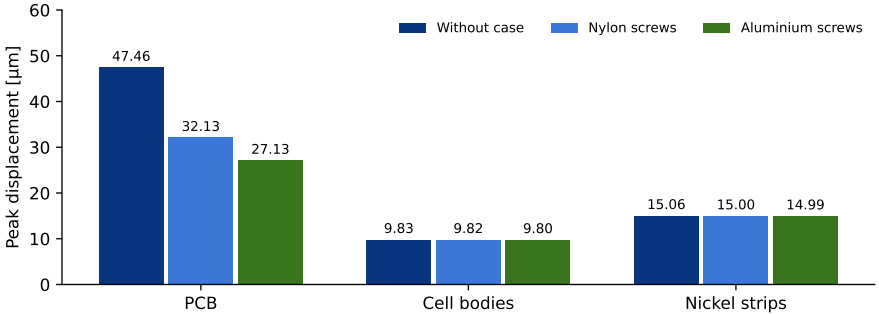
Cell travel fell from 0.50 mm to about 0.21 mm. Peak nickel plastic strain was 1.25% bare; the two cased states bracketing 8.54 N gave 0.36% and 0.43%.

At the same 0.50 mm travel

The pull rose from 8.54 to 138.53 N. The four case screw supports carried 112.15 N. Peak nickel plastic strain rose from 1.25% to 1.51%.

Local fixture, 3 mm stand-off, positive X pull with fixed screw tops.

Aluminium screws further reduced PCB deflection during tightening

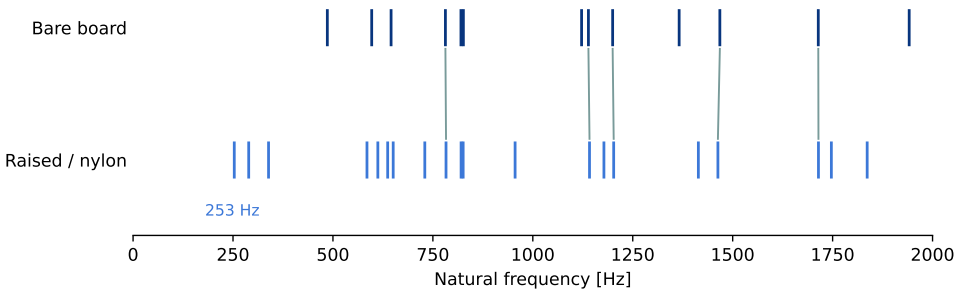


The same PCB node moved 47.46 μm bare, 32.13 μm with nylon screws and **27.13 μm** with aluminium screws. The reductions from bare were 32% and 43%, respectively.

No cell, strip or washer contacted the case during tightening. Load transfer was through the four corner attachments.

3 mm stand-off. Six original bolts loaded to 750 N each and locked. No imposed vibration.

The raised case introduces three modes below 486 Hz

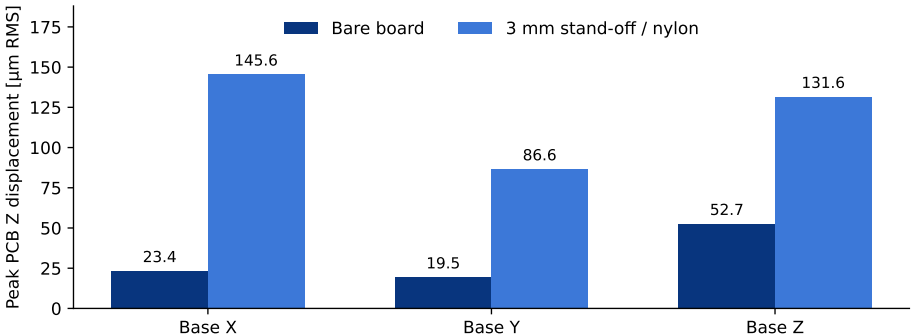


The lowest cased mode is at **253 Hz**; its cell motion peaks at only 2.8% of the PCB peak. The bare board has no close shape match for this mode.

The 781 Hz bare battery-motion mode reappears at 783 Hz with the case (shape correlation 0.943). Comparing modes by rank alone would miss this match.

Bolt-locked flexible assemblies; four nylon case screws at 3 mm stand-off. Undamped mode shapes about the tightened state.

The open-gap vibration model predicts more PCB motion



Under base X, the PCB's largest Z response rises by a factor of **6.23**. The cells' largest X response falls by **12.4%**. These quantities describe different parts and directions.

Native MAPDL confirms the 6.23-fold rise under base X. In the same load case, the peak PCB equivalent stress fell from 90.32 to 88.85 MPa RMS and moved about 54 mm across the board. Peak PCB Y strain more than doubled. The seeded motion also predicts cell contact.